

Milo Shan

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github.com/MoodyMarshmallow

Education

University of Pennsylvania, BS in Artificial Intelligence | GPA 3.6 Aug 2024 – May 2028

- Relevant Courses: Applied Machine Learning, Deep Learning Lab, Human-Computer Interaction

Experience

Pattern, Software Engineering Intern – *Lehi, UT* May 2026 - Aug 2026

- Embedded as an Amazon copywriter and content manager to map end-to-end content workflow; presented findings to Data Science leadership and catalyzed team-wide pivot from maintaining legacy tools to building AI-infrastructure.
- Developed 9 platform-agnostic tools exposing existing SEO optimization pipeline to Claude Cowork and an internal chatbot, cutting end-to-end turnaround from 4–8 weeks to 3–4 days.
- Leading alpha-testing with ~20 users, translating feedback into product requirements for PMs and shipping 10+ fixes across 3 repositories.

Pensieve, AI Engineer (contract) – *San Francisco, CA* Jan 2026

- Designed LLM-based pipeline for AI grading tool that extracted student names and IDs from scanned exams, enabling papers to be matched automatically to the correct student.
- Implemented image pre-processing and fuzzy matching system based on Levenshtein Distance, achieving >80% accuracy over hundreds of papers.

OpenAI, ChatGPT Lab Member – *New York, NY* May 2025 - Present

- Contributed to ChatGPT Study Mode development, serving 700M+ weekly users through feature testing.

UPenn PennNLP, Researcher and Backend Lead – *Philadelphia, PA* May 2025 - Aug 2025

- Led 7-person team in developing a multi-agent interaction platform, with a game-like, turn-based simulation environment for multi-agent interaction research.
- Developed visualization layer in Godot Engine for interactive simulation monitoring and debugging.

Publications

Computational prediction of MOFs with the potential to improve the efficiency of industrial CO₂ capture July 2024

Machine Learning Regression to Predict Soil Moisture in Domestic Garden Environments May 2023

Projects

NanoGPT Training Speedrun [Github link](#)

- Optimized training of nanoGPT on ~900M tokens of FineWeb dataset for speed and accuracy.
- Implemented and tested gated attention variants to an existing speedrun record.
- Achieved 0.6% accuracy improvement.

Fintuning of DINOv3 for Image2GPS Task [Github link](#)

- Designed preprocessing pipeline cleaning ~3,500 self-collected phone-photo metadata rows; delegating them to ~90 contiguous regions via rasterization, k-medoids, and custom growth algorithm.
- Fine-tuned DINOv3-S classifier using soft adjacent-region labels & top-5 weighted-centroid decoding for final lat-long prediction. Achieved 37.27 m external-test error.

Technical Skills

Languages: Python, C++, Java, C#, TypeScript, CSS, SQL

Frameworks & Technologies: PyTorch, React, Next.js, FastAPI, Tailwind CSS, Godot Engine

Tools & Platforms: PostgreSQL, Vite, Supabase, Linux, Jupyter, Arduino, Git

AI & Machine Learning: Machine Learning Model Evaluation, Large Language Models (LLMs) Fine-tuning